Advances in AI-Based Toxicity Prediction for Drug Discovery

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Targeted Research Synthesis for Term Paper

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I. INTRODUCTION

Drug toxicity is a major reason why promising drug candidates fail during development. A compound may treat a disease effectively but still damage the liver, heart, kidneys, nervous system, or DNA. Traditional laboratory and animal tests remain necessary, yet they are expensive, slow, and may not fully predict human reactions. Detecting harmful compounds early is therefore important for patient safety and for reducing wasted time and cost.

Artificial intelligence (AI) supports early toxicity screening by learning patterns from chemical structures and biological data. Traditional machine-learning models use molecular descriptors and fingerprints, while newer deep-learning systems use SMILES strings, molecular graphs, attention mechanisms, and multimodal information. Recent tools predict broad ADMET properties or specific risks such as liver injury, hERG-related heart toxicity, neurotoxicity, mutagenicity, genotoxicity, and carcinogenicity. However, high accuracy on one dataset does not guarantee reliable performance on new compounds. Data imbalance, inconsistent labels, weak external validation, limited biological context, and poor interpretability remain important challenges. For practical use, AI models should provide clear explanations, uncertainty estimates, and warnings when a compound falls outside the model's training domain.

II. STATE-OF-THE-ART (SOTA)

Table I compares twenty recent papers: four broad reviews and sixteen individual prediction platforms or model-development studies. Each work is summarized by toxicity scope, AI approach, reported result, strength, and limitation. Because the studies use different datasets and validation procedures, their scores should be interpreted within each study rather than used as a direct ranking.

TABLE I: STATE-OF-THE-ART COMPARISON OF TWENTY RECENT PAPERS

Paper/Method	Endpoint or Scope	Input and AI Approach	Reported Result	Main Strength	Main Limitation
Lee et al. [1]	Review of multiple toxicity endpoints	Reviews RF, SVM, XGBoost, neural networks, GNNs, transformers, descriptors, and molecular graphs.	No single score	Recent overview of datasets, models, endpoints, evaluation, and interpretability.	Review paper; does not experimentally validate one new model.
Zhang et al. [2]	Acute, organ-specific, genetic, and cancer toxicity	Reviews ML, deep learning, transfer learning, multimodal fusion, databases, and tools.	No single score	Broad framework covering databases, endpoints, challenges, and future directions.	Review comparisons depend on results reported by other studies.
Gianquinto et al. [3]	Integrative drug safety	Reviews QSAR, molecular	No single score	Connects computational	Review paper; no common benchmark

	and predictive toxicology	modelling, ML, multi-omics, toxicokinetics, and advanced in vitro systems.		prediction with mechanisms and clinical translation.	or single model is evaluated.
Ali [4]	AI-driven drug-toxicity prediction	Reviews QSAR, ML, deep learning, toxicity databases, algorithms, and software tools.	No single score	Clear summary of approaches, challenges, regulation, and future opportunities.	Limited original quantitative comparison or external validation.
ADMETLab 3.0 [5]	Broad ADMET and toxicity	Molecular graphs with directed message-passing neural networks.	hERG AUROC about 0.94	Predicts 119 endpoints and provides uncertainty and structural alerts.	Benchmark results do not confirm performance in clinical settings.
ProTox 3.0 [6]	Organ, clinical, and general toxicity	ML, chemical similarity, molecular targets, initiating events, and adverse outcome pathways.	Endpoint-dependent	Broad coverage with some mechanistic information.	Dataset size, quality, and performance vary across endpoints.
Deep-PK [7]	35 toxicity and pharmacokinetic endpoints	Graph neural network using molecular structures and physicochemical descriptors.	Endpoint-dependent	Highlights molecular subgraphs that influence predictions.	Limited dose, exposure, patient, and clinical information.
InterDILI [8]	Drug-induced liver injury	Fingerprints and descriptors with RF, LightGBM, logistic regression, and neural networks.	AUROC 0.97; AUPRC 0.95; accuracy 0.90	Strong performance with feature and attention explanations.	Performance decreased under a different data split.
DILI Predictor [9]	Drug-induced liver injury	Fingerprints, descriptors, predicted pharmacokinetic properties, proxy labels, and random forest.	AUROC 0.63	Combines predicted and experimental information in an accessible tool.	Moderate discrimination and a narrow endpoint.
GeoDILI [10]	Drug-induced liver injury	Three-dimensional molecular graphs with a geometry-based graph neural network.	AUROC 0.908; accuracy 0.975	Atom-level explanations of possible toxic molecular regions.	Needs stronger independent, temporal, and prospective validation.
OvA-QSTR [11]	Drug-induced liver injury	PaDEL descriptors with Bayesian networks, decision trees, and random forests.	AUPRC 0.718-0.869	Relatively simple and understandable machine-learning models.	Hand-crafted descriptors may miss complex mechanisms.
Rao et al. [12]	Severity of drug-induced liver injury	Physicochemical descriptors and off-target profiles	AUROC 0.88;	Predicts severity and includes	Depends on the availability and

		with RF, SVM, neural networks, and logistic regression.	specificity 0.90	off-target information.	accuracy of curated off-target data.
hERGBoost [13]	hERG inhibition and heart toxicity	Molecular descriptors and fingerprints with XGBoost classification and regression.	Accuracy 0.814; MCC 0.614; R2 0.622	Predicts both toxicity class and continuous IC50 values.	Limited explanation of why a molecule is risky.
hERGAT [14]	hERG inhibition	Molecular graphs, fingerprints, descriptors, graph attention networks, and gated recurrent units.	AUROC 0.907; AUPRC 0.904	Atom-level and molecule-level attention explanations.	hERG inhibition is only one mechanism of cardiotoxicity.
AttenhERG [15]	hERG inhibition	Molecular graph representation with an AttentiveFP graph neural network.	AUROC 0.835; MCC 0.543	Provides uncertainty and atom-level attention.	Lower discrimination than several other hERG models.
DMFGAM [16]	hERG inhibition	Extended-connectivity and atom-pair fingerprints combined with graph attention networks.	AUROC 0.894; accuracy 0.817	Combines molecular-graph and fingerprint information.	Does not include clinical, metabolic, dose, or omics data.
NeuTox 2.0 [17]	BBB effects, neuronal damage, and neurotoxicity	Molecular graphs, fingerprints, descriptors, DMPNN, MFBERT, and multimodal feature fusion.	AUROC up to 0.971	Predicts several related neurotoxicity outcomes.	Most input features are still structure-derived.
muTOX-AL [18]	Ames mutagenicity	ECFP fingerprints, MACCS keys, descriptors, 2D-CNN, and active learning.	AUROC 0.909	Strong performance with fewer labelled training samples.	Requires repeated expert or laboratory annotation.
Fournier et al. [19]	Genotoxicity across several assays	2D descriptors and MACCS keys with similarity search and extremely randomized trees.	AUROC about 0.921-0.972	Covers several genotoxicity assays.	No complete executable prediction system was reported.
DCAMCP [20]	Carcinogenicity	Multiple molecular fingerprints with capsule networks and self-attention routing.	AUROC 0.793; accuracy 0.718	Uses attention and reports balanced classification measures.	Moderate performance and limited carcinogenicity data.

Note: The first four rows are review papers; the remaining sixteen rows describe individual platforms or model-development studies. AUROC and AUPRC values closer to 1 indicate stronger discrimination. Accuracy is the proportion of correct predictions, and MCC ranges from -1 to 1. Results from different datasets, endpoints, and validation procedures are not directly comparable

III. CONCLUSION

AI-based toxicity prediction has progressed from descriptor-based machine learning to graph neural networks, attention, active learning, and multimodal systems. These tools can identify possible safety risks earlier, but no single model is reliable for every endpoint. Current limitations include inconsistent datasets, weak generalization, limited biological information, and unclear model decisions. Future systems should combine molecular structure with dose, metabolism, off-target effects, and biological data while providing uncertainty and applicability warnings. AI should support, not replace, laboratory testing and clinical judgement. With transparent validation and human oversight, it can help reduce late-stage failures and support safer drug development.

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